

UCHEOMA OKOMA

Cybersecurity | Cloud Security | Healthcare Security
Claremont, CA (USA) • [Email](#) • [LinkedIn](#) • [Medium](#) • [Portfolio](#)

SUMMARY

Cybersecurity master's student with a medical background and experience in API security, cloud security, AI privacy, and healthcare systems. Hands-on experience assessing vulnerabilities, building security-focused solutions, and applying threat modeling and risk assessment frameworks in technical environments.

EDUCATION

Claremont Graduate University M.S. Information Systems & Technology <i>Claremont, USA.</i>	2025 – Present 4.0 GPA
Seneca College Post-Graduate Certificate, Cybersecurity & Threat Management <i>Ontario, Canada</i>	2025 3.8 GPA
V.N. Karazin Kharkiv National University Doctor of Medicine (MD)	2017 – 2024

EXPERIENCE

Research Assistant — Health Informatics & Security | Global Health Intelligence (GHI) 2026 – Present

- Assess security considerations of healthcare data systems being consolidated into an interoperable platform by mapping data flows and identifying trust boundaries.
- Apply UMLS, HIPAA, and STRIDE threat-modeling principles to support security recommendations.
- Contribute to risk assessments through reports, data-flow diagrams, and decision trees for technical and non-technical stakeholders.

Software Engineer Intern — INSPIRE UVic | Victoria, B.C Sept. 2023 – Dec. 2023

- Contributed to the development of a computer-vision solution for wildfire smoke detection and trespass monitoring using existing camera infrastructure supporting monitoring of regional watershed and rainforest areas.
- Supported model integration and application development using YOLOv8, OpenCV, React, Node.js, and MySQL.
- Collaborated with researchers, software developers, and community stakeholders to test and refine system functionality for real-world environmental monitoring use cases.

SECURITY PROJECTS

API Security Assessment — vAPI Lab | APISEC University (12-week Bootcamp) 2026

- Assessed a vulnerable API against the OWASP API Top 10, identifying vulnerabilities including Broken Object Level Authorization (BOLA), credential stuffing, and Broken Function Level Authorization (BFLA).
- Used Burp Suite, Postman, and ffuf to validate vulnerabilities and document reproducible attack paths.
- Produced technical documentation outlining impact, evidence, and remediation recommendations.

IoT Security Monitoring with AWS, Suricata & ELK Stack

- Built an end-to-end IoT traffic monitoring environment using AWS IoT Core, EC2, Suricata IDS, Filebeat, Elasticsearch, and Kibana.
- Simulated abnormal IoT traffic patterns using Python to evaluate detection capabilities and validate alert generation.
- Configured log forwarding pipelines from Suricata through Filebeat into Elasticsearch and developed Kibana dashboards to visualize security events and trends.
- Troubleshoot authentication, logging, and alert-threshold issues while improving visibility into cloud-connected IoT traffic.

AI Privacy-Aware Chatbot | Python, FastAPI, Claude API

- Developed a privacy-focused AI chatbot using Python, FastAPI, and Tkinter that redacts sensitive information before forwarding prompts to a third-party LLM.
- Implemented pattern-based detection and masking of emails, passwords, file paths, and hashes using regular expressions and custom redaction logic.
- Built an anonymous session-logging system with unique session identifiers and redacted audit logs to support privacy-conscious monitoring.

- Integrated Anthropic Claude through the OpenRouter API while maintaining separation between user inputs, redaction workflows, and model interactions.

TECHNICAL SKILLS

Security & Assessment: OWASP Top 10, OWASP API Top 10, STRIDE, MITRE ATT&CK, NIST CSF, NIST AI RMF, Threat Modeling, Risk Assessment, HIPAA

Cloud & Infrastructure: AWS, Docker, Elasticsearch, Kibana, Suricata IDS, AWS IoT Core

Tools: Burp Suite, Postman, ffuf, Nmap, Wireshark, Metasploit

Programming: Python, SQL, JavaScript, Java, Bash

Certifications: CompTIA Security+, AWS Cloud Practitioner, AWS Cloud Security Foundations, CyberSafe API Security Certification.